

LAB EXPERIMENTS (16-20)

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Set 16:

Customer Demographics Analysis

Dataset: Customer Demographics

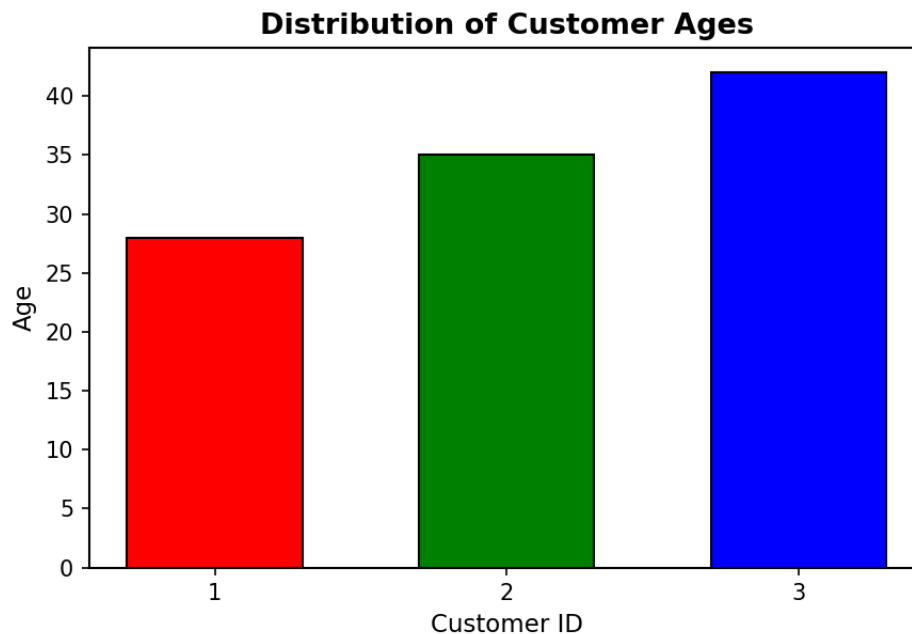
1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

CODE:

```
# Dataset
customer_id <- c(1, 2, 3)
age <- c(28, 35, 42)
gender <- c("Female", "Male", "Female")
income <- c(50000, 60000, 75000)

# Bar Chart
barplot(age,
  names.arg = customer_id,
  col = c("red", "green", "blue"),
  xlab = "Customer ID",
  ylab = "Age",
  main = "Distribution of Customer Ages")
```

OUTPUT:



2. Generate a pie chart to represent the distribution of customers by gender.

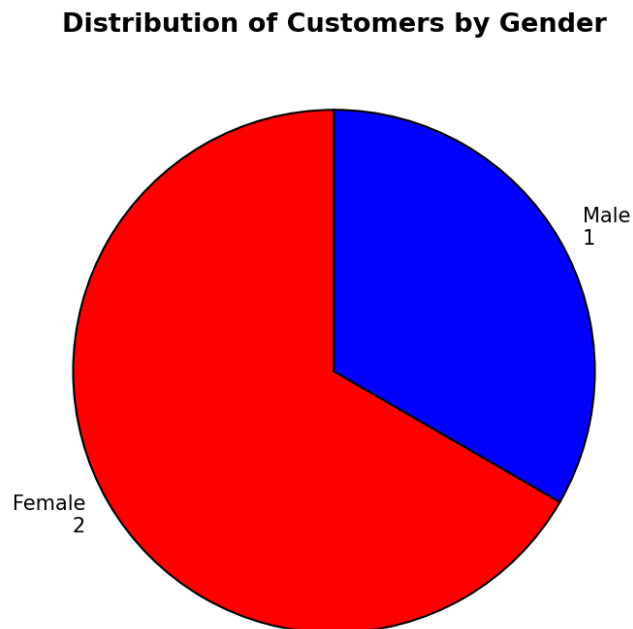
CODE:

```
# Dataset
gender <- c("Female", "Male", "Female")

# Count by gender
gender_count <- table(gender)

# Pie Chart
pie(gender_count,
  labels = paste(names(gender_count), gender_count, sep = "\n"),
  col = c("red", "blue"),
  main = "Distribution of Customers by Gender")
```

OUTPUT:



3. Build a table to show the demographic information for each customer. Label the table elements.

CODE:

```
# Dataset
customer_data <- data.frame(
  Customer_ID = c(1, 2, 3),
  Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)
```

```
# Display Table
print(customer_data)
```

OUTPUT:

	Customer_ID	Age	Gender	Income
1	1	28	Female	50000
2	2	35	Male	60000
3	3	42	Female	75000

Set 17:

Employee Performance Analysis

Dataset: Employee Performance

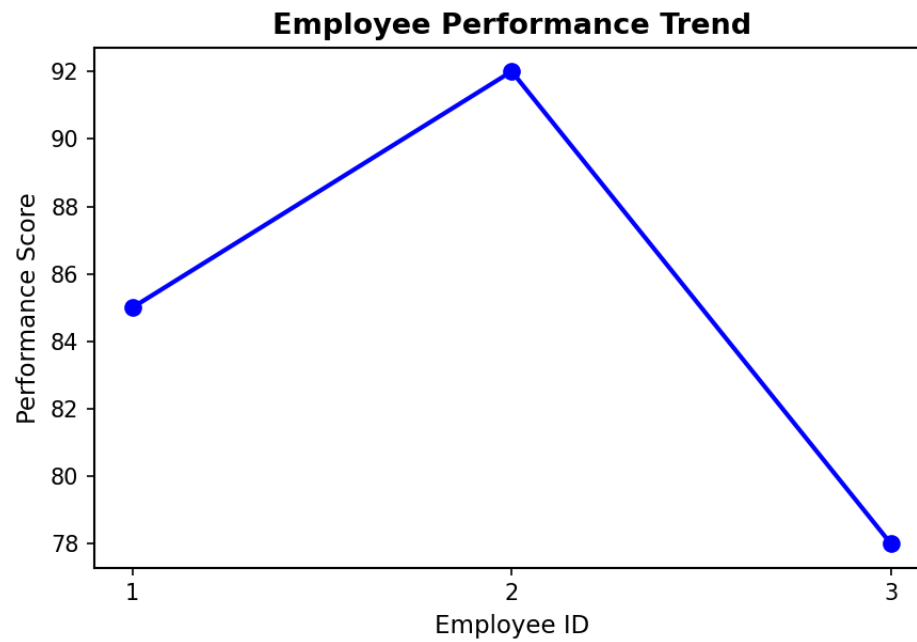
1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

CODE:

```
# Dataset
employee_id <- c(1, 2, 3)
department <- c("Sales", "HR", "Marketing")
years_of_service <- c(5, 3, 7)
performance_score <- c(85, 92, 78)

# Line Chart
plot(employee_id,
      performance_score,
      type = "o",
      col = "blue",
      pch = 16,
      lwd = 2,
      xlab = "Employee ID",
      ylab = "Performance Score",
      main = "Employee Performance Trend")
```

OUTPUT:



2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

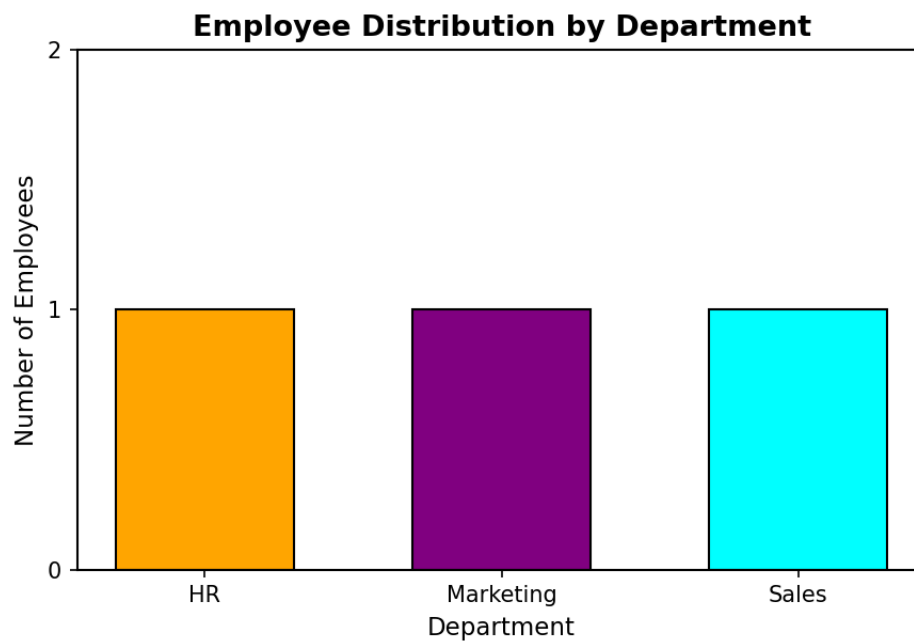
CODE:

```
# Dataset
department <- c("Sales", "HR", "Marketing")

# Count employees per department
dept_count <- table(department)

# Bar Chart
barplot(dept_count,
        col = c("orange", "purple", "cyan"),
        xlab = "Department",
        ylab = "Number of Employees",
        main = "Employee Distribution by Department")
```

OUTPUT:



3. Build a table to display the performance data for each employee. Label the table elements.

CODE:

```
# Dataset
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)

# Display Table
print(employee_data)
```

OUTPUT:

	Employee_ID	Department	Years_of_Service	Performance_Score
1	1	Sales	5	85
2	2	HR	3	92
3	3	Marketing	7	78

Set 18:

Product Inventory Management

Dataset: Product Inventory

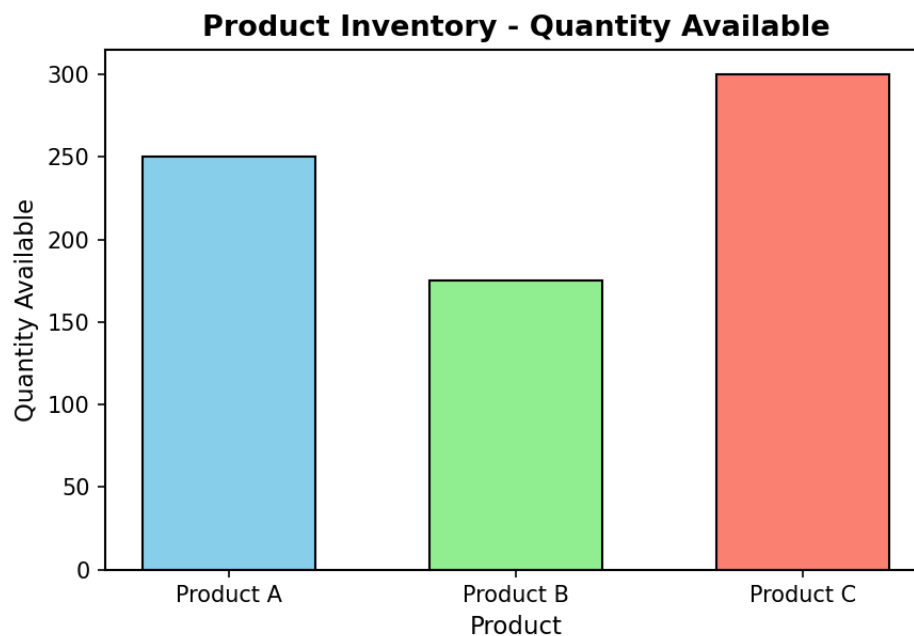
1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

CODE:

```
# Dataset
product <- c("Product A", "Product B", "Product C")
quantity <- c(250, 175, 300)
price <- c(20, 15, 18)

# Bar Chart
barplot(quantity,
  names.arg = product,
  col = c("skyblue", "lightgreen", "salmon"),
  xlab = "Product",
  ylab = "Quantity Available",
  main = "Product Inventory - Quantity Available")
```

OUTPUT:



2. Generate a stacked bar chart to show the quantity of each product within different product categories.

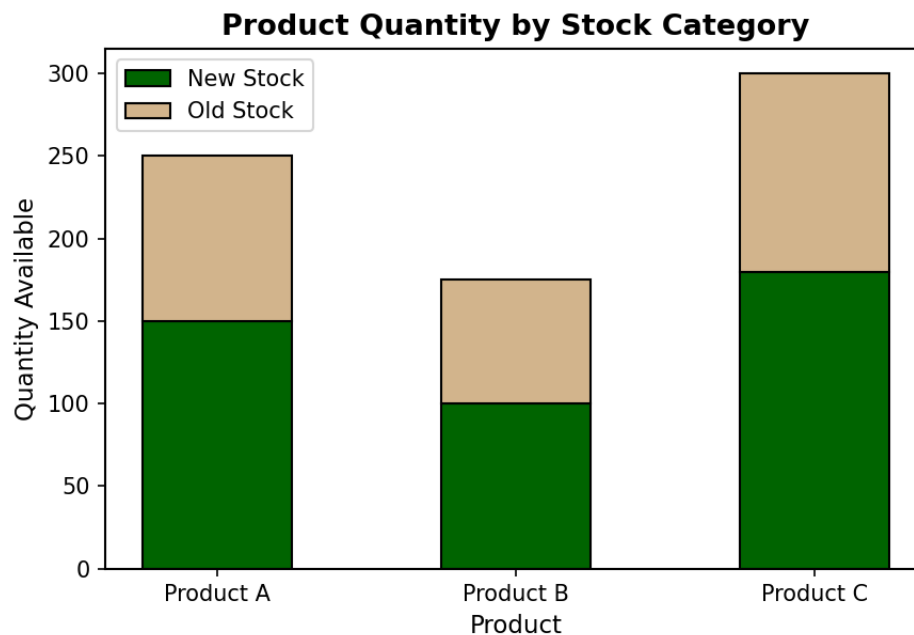
CODE:

```
# Dataset
product <- c("Product A", "Product B", "Product C")
new_stock <- c(150, 100, 180)
old_stock <- c(100, 75, 120)

# Create matrix (categories as rows)
qty_matrix <- rbind(new_stock, old_stock)
colnames(qty_matrix) <- product

# Stacked Bar Chart
barplot(qty_matrix,
  col = c("darkgreen", "tan"),
  xlab = "Product",
  ylab = "Quantity Available",
  main = "Product Quantity by Stock Category")
legend("topright",
  legend = c("New Stock", "Old Stock"),
  fill = c("darkgreen", "tan"))
```

OUTPUT:



3. Build a table to show the inventory data for each product. Label the table elements.

CODE:

```
# Dataset
inventory_data <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  Quantity_Available = c(250, 175, 300),
  Price = c(20, 15, 18)
)

# Display Table
print(inventory_data)
```

OUTPUT:

	Product_ID	Product_Name	Quantity_Available	Price
1	1	Product A	250	20
2	2	Product B	175	15
3	3	Product C	300	18

Set 19:

Survey Responses Analysis

Dataset: Survey Responses

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

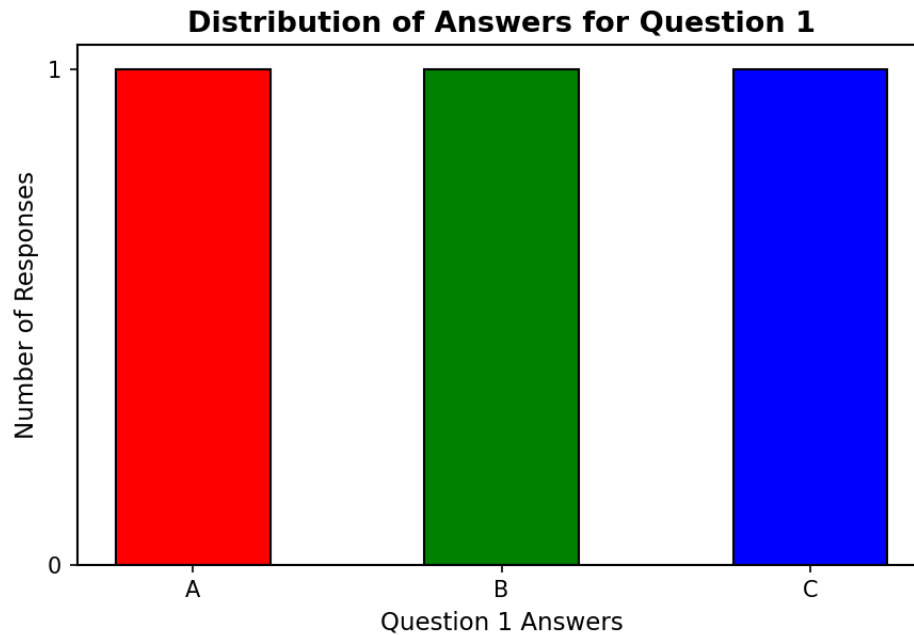
CODE:

```
# Dataset
Q1 <- c("A", "B", "C")

# Count responses
q1_count <- table(Q1)

# Grouped Bar Chart
barplot(q1_count,
  col = c("red", "green", "blue"),
  xlab = "Question 1 Answers",
  ylab = "Number of Responses",
  main = "Distribution of Answers for Question 1")
```


OUTPUT:



2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

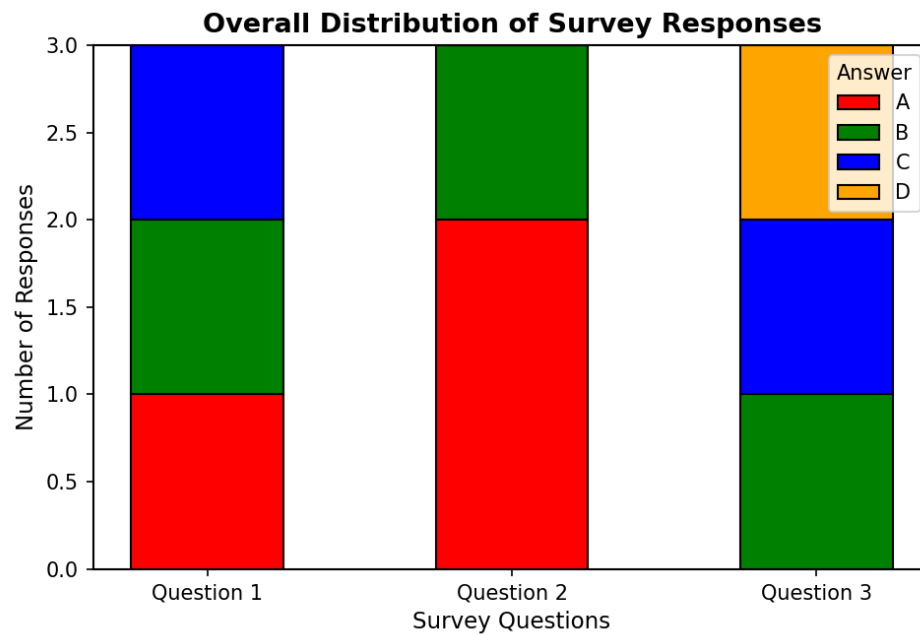
CODE:

```
# Dataset
Q1 <- c("A", "B", "C")
Q2 <- c("B", "A", "A")
Q3 <- c("C", "D", "B")

# Build frequency matrix (answers x questions)
answers <- c("A", "B", "C", "D")
resp_matrix <- sapply(list(Q1, Q2, Q3), function(q) table(factor(q, levels = answers)))
colnames(resp_matrix) <- c("Question 1", "Question 2", "Question 3")

# Stacked Bar Chart
barplot(resp_matrix,
        col = c("red", "green", "blue", "orange"),
        xlab = "Survey Questions",
        ylab = "Number of Responses",
        main = "Overall Distribution of Survey Responses")
legend("topright", legend = answers,
      fill = c("red", "green", "blue", "orange"), title = "Answer")
```

OUTPUT:



3. Build a table to show the survey response data for each survey. Label the table elements.

CODE:

```
# Dataset
survey_data <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)

# Display Table
print(survey_data)
```

OUTPUT:

	Survey_ID	Question1	Question2	Question3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B

Set 20:

Stock Analysis

Dataset: Stock Prices

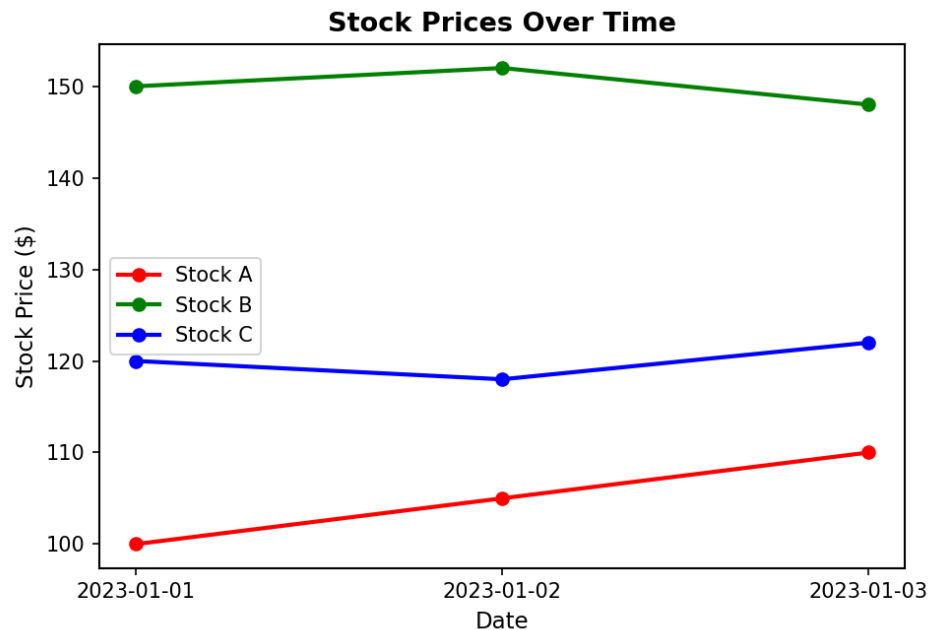
1. Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock C) over a specific time period. Label the axes and title the chart.

CODE:

```
# Dataset
date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03"))
stock_A <- c(100, 105, 110)
stock_B <- c(150, 152, 148)
stock_C <- c(120, 118, 122)

# Line Chart
plot(date, stock_A,
     type = "o", col = "red", lwd = 2, pch = 16,
     ylim = range(c(stock_A, stock_B, stock_C)),
     xlab = "Date", ylab = "Stock Price ($)",
     main = "Stock Prices Over Time")
lines(date, stock_B, type = "o", col = "green", lwd = 2, pch = 16)
lines(date, stock_C, type = "o", col = "blue", lwd = 2, pch = 16)
legend("topright",
     legend = c("Stock A", "Stock B", "Stock C"),
     col = c("red", "green", "blue"), lwd = 2)
```

OUTPUT:



2. Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.

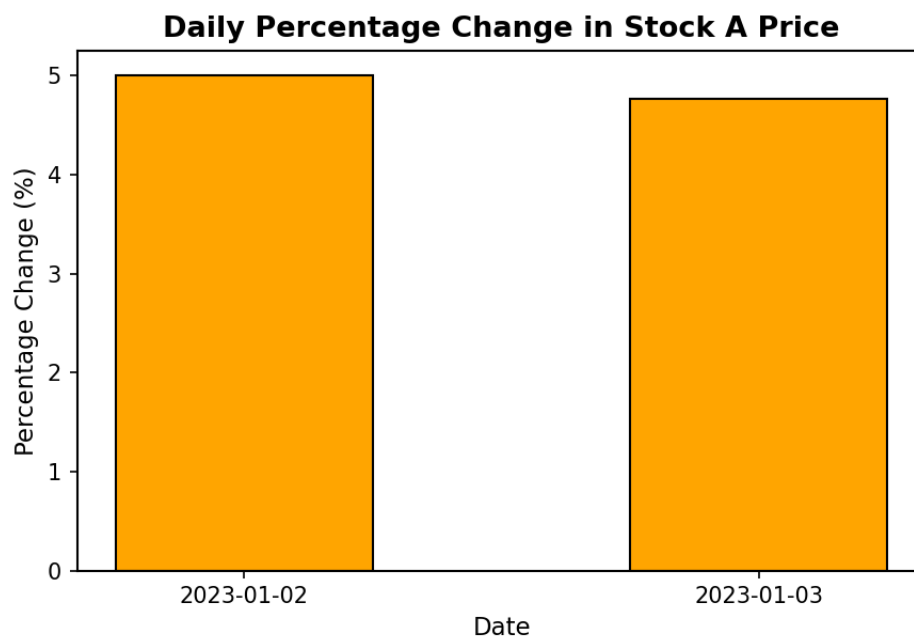
CODE:

```
# Dataset
date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03"))
stock_A <- c(100, 105, 110)

# Calculate daily percentage change
pct_change <- round(diff(stock_A) / head(stock_A, -1) * 100, 2)

# Bar Chart
barplot(pct_change,
        names.arg = as.character(date[-1]),
        col = "orange",
        xlab = "Date",
        ylab = "Percentage Change (%)",
        main = "Daily Percentage Change in Stock A Price")
```

OUTPUT:



3. Build a table to display the stock price data for each company over the given period. Label the table elements.

CODE:

```
# Dataset
stock_data <- data.frame(
  Date = c("2023-01-01", "2023-01-02", "2023-01-03"),
  Stock_A = c(100, 105, 110),
```

```
    Stock_B = c(150, 152, 148),  
    Stock_C = c(120, 118, 122)  
  )  
  
# Display Table  
print(stock_data)
```

OUTPUT:

	Date	Stock_A	Stock_B	Stock_C
1	2023-01-01	100	150	120
2	2023-01-02	105	152	118
3	2023-01-03	110	148	122